

SEMESTER IV

Course Code	Course name	L	T	P	C
CSEG2065	Data communication and Networks	3	0	1	4
Total Units to be Covered: 5		Total Contact Hours: 75			
Prerequisite(s):		Syllabus version: 1.0			

Course Objectives

The objectives of this course are as follows:

- 1) Understand the basic components and functions of computer networks, including network topologies, protocols, and networking devices.
- 2) Understand need of layered architecture and differentiate OSI and TCP/IP
- 3) Gain an understanding of error and flow control techniques on communication channels.
- 4) Explore routing algorithms and its application.
- 5) Get a brief idea about network analysis tools (Wireshark, NMAP).

Course Outcomes

The outcomes of this course are as follows:

CO1: Evaluate network devices functionality and network command significance.

CO2: Evaluate and address problems of error control, flow control, and channel access.

CO3: Analyze and adopt fundamental workings of routing algorithms.

CO4: Create solutions for recent challenges in large-scale networks.

CO5: Apply knowledge of network traffic analysis tool to investigate network activities.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	2	2	2	3	-	-	-	-	2	-	-	3	-	-	-
CO 2	-	2	3	2	-	-	-	-	-	-	-	3	2	-	-
CO 3	2	3	3	3	2	2	-	-	2	-	-	3	3	2	-
CO4	3	3	3	3	2	-	-	2	2	-	2	3	2	3	-

CO5	2	3	3	3	3	3	-	2	2	2	3	3	3	3	-
Average	1.8	2.6	2.8	2.8	1.4	1	-	0.8	1.6	0.4	1	3	2	1.6	-

1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

“ - ” means there is no correlation

Syllabus

Unit I: Basic of Computer Networking and Technologies 9 Lecture Hours

Introduction to Computer Networking Concepts: Layered Network Protocol Architectures (OSI, TCP/IP); LAN, WAN, MAN, PAN, LAN Topologies; Connectivity Devices and Cable Types; Ethernet, Gigabit Ethernet (GbE); Circuit Switching, Message Switching, and Packet Switching; WiMAX, 5G and beyond, Cellular Technology, Communication Channels and performance metrics.

Unit II: Data Link Layer - Part 1 9 Lecture Hours

Logical Link Control (LLC) sub-layer: Framing, Data Communication Character Codes, Error Control: Error Detection (Redundancy Checking: VRC, Checksum, LRC, CRC); Retransmission, Error Correction: Forward Error Correction (Hamming Code), Character Synchronization, Reliable transmission and Automatic Repeat Request (ARQ) protocols including Stop-and-Wait, Go-back-N, Selective Repeat; Performance analysis of ARQ protocols; Example protocols such as HDLC and PPP.

Unit III: Data Link Layer - Part 2 8 Lecture Hours

Medium Access Control (MAC) sub-layer: Channel Allocation Problems, Multiple Access Protocols and Types: TDMA, FDMA, CSMA, CSMA/CD, CSMA/CA protocols; Hidden Node and Exposed Node Problems, Performance analysis; Shared and Switched Ethernet; IEEE Standards 802.3 & 802.11, 10-Gigabit Ethernet.

Unit IV: Network Layer 10 Lecture Hours

Network Layer Design Issues, Network Address Translation, Internet Protocol (IP): IPv4 and IPv6 addressing; IP Addressing Techniques: Classful Addressing, Classless Addressing, Network and Host Identification, Loopback Address, Broadcast Address, Address Masking; Networks and Subnetworks: Subnetting, Subnet Mask, Supernetting; Network-Layer Protocols: ARP, RARP, IP datagram; Internetworking: Routing and Routing protocols (distance-vector and link-state); Interior and Exterior Gateway Protocol concepts; Routing Algorithms including Dijkstra's algorithm and distributed Bellman-Ford algorithm; Example protocols: OSPF, RIP, BGP, Encapsulation and Tunneling, Congestion Control, Quality of Service, Introduction of Wireshark Tool.

Unit V: Transport Layer

9 Lecture Hours

Introduction and Transport-Layer Services, Port Address, Socket Address; Internet Transport Protocols: UDP, Introduction to UDP, Remote Procedure call, Real-time Transport Protocols; Internet Transport Protocols: TCP, service model, TCP protocol, TCP segment header, TCP Connection establishment, TCP Connection Release, TCP Connection management modeling, TCP sliding window, TCP Timer management, TCP Congestion control; Performance Issues: Performance problems in computer networks, Network Performance Management, Host Design for fast networks; Fast segment processing, Header compression, protocols for long Fat networks; Virtual Private Network (VPN); Introduction of Nmap Tool.

Total lecture Hours 45

References*

Textbooks	1. James F. Kurose, and Keith W. Ross, "Computer Networking : A Top-Down Approach", 8th Edition, Pearson, 2022. 2. W. Tomasi, "Introduction to data communications and networking", 5th edition, Prentice-Hall, Inc., 2008.
Reference books	1. Walter Goralski, "The illustrated network: how TCP/IP works in a modern network", 2nd Edition, Morgan Kaufmann, 2017. 2. Andrew S. Tanenbaum, "Computer Networks", 5th

	Edition, Pearson Education, 2023. 3. L. L. Peterson, and B. S. Davie, "Computer networks: a systems approach", 6th Edition, Elsevier, 2020.
--	--

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination etc.

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100



Data communication and Networks Lab

List of Experiments

Experiment 1: Familiarization with networking devices. **(CO1)**

Experiment 2: Write a program for bit stuffing and de-stuffing in a bit stream. **(CO2)**

Experiment 3: Write a program for CRC and Hamming Code. **(CO2)**

Experiment 4: Familiarization with Network IP, subnetting and supernetting. **(CO3)**

Experiment 5: Familiarization of basic network command and network configuration commands. **(CO1, CO5)**

Experiment 6: Set up a network topology in Cisco Packet Tracer (Ring, Bus, Star, Mesh etc.) **(CO4, CO5)**

Experiment 7: Set up network topology in two and more than two routers. **(CO4, CO5)**

Experiment 8: Distance vector routing protocol **(CO3)**

Experiment 9: Link-state vector routing protocol **(CO3)**

Experiment 10: Familiarization with network monitoring tools (NMAP and Wireshark) **(CO5)**

Experiment 11: Capture network traffic using Wireshark. **(CO5)**

Experiment 12: Analyzing network traffic using Wireshark. **(CO5)**

Total Lab hours 30

References*

Textbooks	1. James F. Kurose, and Keith W. Ross, "Computer Networking : A Top-Down Approach", 8th Edition, Pearson, 2022. 2. Andrew S. Tanenbaum, "Computer Networks", 5th Edition, Pearson Education, 2023.
Reference books	1. W. Tomasi, "Introduction to data communications and networking", 5th edition, Prentice-Hall, Inc., 2008.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination etc.

Examination Scheme: Continuous Assessment

Components	Quiz & Viva	Performance & Lab Report
Weightage (%)	50	50

